

Data Control Language (DCL) Commands

Q. What is Data Control Language (DCL) Explain the concept of user privileges and demonstrate the use of GRANT and REVOKE commands.

[DEF] Definition to Remember

Data Control Language (DCL) is a subset of SQL used by Database Administrators (DBAs) to control user access and maintain database security. It contains two commands: GRANT (give permissions) and REVOKE (remove permissions). DCL enforces the Principle of Least Privilege.

1. User Privileges and Access Control

A Privilege is a specific permission to perform an action on a database object.

Privilege Type	Description	Examples
System Privileges	Administrative tasks	`CREATE TABLE`, `CREATE USER`
Object Privileges	Actions on specific tables/views	`SELECT`, `INSERT`, `UPDATE`, `DELETE`

2. GRANT Command

Gives specific privileges to a user or role.

Syntax:

* WITH GRANT OPTION allows the receiving user to further grant the privilege to others.

Examples:

```
GRANT
```

```
-- Gra
GRANT
-- Gra
GRANT
```

3. REVOKE Command

Removes previously granted privileges from a user or role.

Syntax:

Examples:

```
REVOKE
```

```
-- Rem
REVOKE
-- Rem
REVOKE
```

[MUST-WRITE] Must-Write Points (for 10 marks)

1. DCL controls who can access which database objects and what actions they can perform.
2. Two DCL commands: GRANT (give privileges) and REVOKE (remove privileges).
3. Two types of privileges: System Privileges (admin tasks) and Object Privileges (table actions).
4. GRANT privilege ON object TO user -- grants specific access.
5. WITH GRANT OPTION allows the user to further pass the privilege to others.
6. REVOKE privilege ON object FROM user -- removes a specific privilege.
7. DCL enforces the Principle of Least Privilege -- give only the access needed.

[RECALL] Quick Recall

DCL -> Security Control -> Privileges (System + Object) -> GRANT (give access, WITH GRANT OPTION) -> REVOKE (remove access) -> Principle of Least Privilege